

Tejashwini Saravanan

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SUMMARY

Clinical Data Analyst with experience in real-world data analytics, clinical trial optimization, and HIPAA-compliant health data platforms. Direct experience building patient screening pipelines, EHR/FHIR data architectures, and population health analytics systems processing **500K+ patient records**. Proficient in **Python, PySpark, SQL, GCP, BigQuery, and Power BI** - with focused expertise in patient risk modeling, real-world evidence generation, and longitudinal EHR data analysis. Passionate about accelerating neurological clinical trial recruitment through data-driven patient identification and scalable health data infrastructure.

EXPERIENCE

Data Analyst | Community Dreams Foundation | Remote Nov 2025 – Present

- Architected end-to-end ETL pipelines and agent orchestration workflows on Databricks as engineering lead of a 4-person team, reducing patient data processing time from **3–5 days to near real-time**, improving data accuracy by **30%**, and enabling HIPAA-compliant ingestion of **500K+ genomic and wearable patient records** - directly supporting clinical decision-making workflows.
- Designed centralized MySQL data models and Power BI dashboards adopted by **5–10 clinical stakeholders across 2 teams**, surfacing real-time population health risk indicators that replaced manual reporting cycles and enabled same-day clinical visibility.
- Automated synthetic PHI-safe dataset generation pipelines for secure model testing, cutting ML release cycles by **~1 week per iteration (25% faster)** - accelerating development of patient risk scoring models for clinical deployment.
- Developed an FDA Safety Alert & Recall Intelligence System ingesting **100–500+ monthly enforcement records** via OpenFDA API, auto-classifying recalls by **FDA Class I/II/III** severity tiers and reducing manual surveillance effort by **30%** through real-time automated alerting.

Data Analyst - Disaster Workforce Engagement | American Red Cross, Northwest Region | Everett, WA Aug 2025 – Oct 2025

- Tracked and analyzed volunteer availability and deployment data across **200+ disaster responders in Snohomish County**, building structured workforce datasets that enabled faster volunteer mobilization for home fire and emergency response operations averaging nearly **3 incidents per day** across the Northwest Region.
- Built leadership reports and dashboards surfacing real-time workforce capacity and deployment readiness, replacing manual data tracking processes and saving **5–7 hours per week** of operational reporting time - enabling Red Cross leadership to make faster staffing decisions during active disaster response windows.
- Analyzed workforce health and capacity patterns across disaster response teams in Snohomish County, identifying volunteer availability trends and risk factors that supported proactive staffing planning - directly applicable to patient cohort availability analysis for **neurological clinical trial recruitment pipelines**.

Data Analyst | Seattle Pacific University, operated by Sodexo | Seattle, WA Oct 2023 – June 2025

- Built interactive Power BI dashboards used daily by **2 operations managers** to track demand patterns, utilization, and performance metrics - contributing to a **5% increase** in customer satisfaction scores through data-backed operational decisions.
- Streamlined reporting by developing automated SQL and Excel workflows, saving **4–6 hours per week** of analyst time and enabling self-serve data access across **2 locations**.

SKILLS

Clinical & RWD/RWE: Clinical Trial Analytics · Patient Screening & Cohort Selection · EHR/FHIR Integration · Real-World Evidence · HIPAA Compliance · Population Health

Programming & Cloud: Python · SQL · PySpark · R · GCP · BigQuery · Vertex AI · Databricks · AWS S3 · Pandas · Scikit-Learn

BI & Visualization: Power BI · Tableau · Looker Studio · Plotly Dash · R Shiny · Flourish Studio

Machine Learning: Random Forest · Gradient Boosting · Neural Networks · K-Means · ROC Analysis · Predictive Risk Modeling

Certifications: Microsoft Office Specialist: Excel Associate (Score: 925/1000)

PROJECTS

Clinical Trial Patient Selection & Admission Prediction | Orange Data Mining · Neural Networks · Gradient Boosting · ROC Analysis | [GitHub](#)

Built a clinical data mining pipeline on **10,000 synthetic patient records** to predict hospital admission type as a patient stability proxy for trial screening - directly modeling the patient recruitment challenge NeuroDiscovery AI solves at scale; ROC analysis established that **longitudinal EHR signals** are required for deployable screening models, informing a companion real-time GCP architecture.

Scalable Health Data Platform: PySpark ML & GCP Architecture | PySpark · GCP · BigQuery · Vertex AI · Cloud Healthcare API · FHIR | [GitHub](#)

Designed a scalable GCP architecture for real-time patient risk scoring at clinical trial network scale, integrating **BigQuery** for RWD analytics, **Vertex AI** for ML model serving, **Cloud Healthcare API** for FHIR-compliant EHR ingestion, and **PySpark streaming pipelines** - mirroring the real-world data infrastructure that powers neurology-specialized patient recruitment platforms like NeuroDiscovery AI.

EDUCATION

Master of Science in Data Analytics for Business · Seattle Pacific University, Seattle WA

Sep 2023 – Aug 2025

Bachelor of Design – Interior Design · SRM Institute of Science and Technology, Chennai, India

June 2018 – May 2022